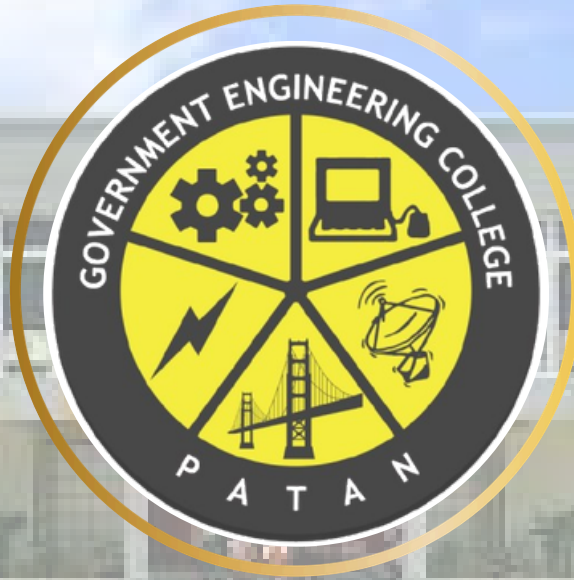




GOVERNMENT ENGINEERING COLLEGE PATAN

NEWSLETTER

INSTITUTE GLIMPSE

Government Engineering College, Patan (G.E.C. Patan), situated in the tranquil and scenic village of Katpur in Patan District, Gujarat, was established in 2004 with a commitment to delivering quality education in engineering and technology. Over the years, the institute has achieved remarkable growth and excellence in academics, innovation, and research. It is affiliated with the Gujarat Technological University (GTU) and functions under the administration of the Directorate of Technical Education (DTE), Government of Gujarat.

INSTITUTE VISION

“To prepare human resources with value-based competency for technical advancements and growth of society.”

INSTITUTE MISSION

- **To Deliver technical programs and services to cater the current needs of society and industry.**
- **Helping industries in solving challenges by means of providing best technical human resources.**
- **To contribute in sustainable growth of society.**



ELECTRONICS AND COMMUNICATION DEPARTMENT

NEWSLETTER

ABOUT

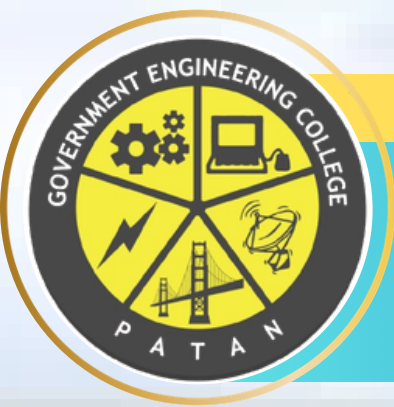
The Electronics and Communication Engineering (ECE) Department at Government Engineering College, Patan, established in 2004, offers a four-year undergraduate program with modern infrastructure, advanced laboratories, and experienced faculty. The department provides strong foundations in electronics, communication, networking, wireless technologies, and VLSI, preparing students for careers, research, and higher education.

DEPARTMENT VISION

“To prepare engineers with essential technical knowledge, value-based professional skills for technical upgradation and societal growth.”

DEPARTMENT MISSION

- **To prepare competent engineers with the ability to solve industrial problems who can raise the living standards of the society.**
- **To prepare engineers with effective communication skills, professional ethics and leadership qualities.**



NEWSLETTER

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HOD'S MESSAGE



It is my immense pleasure to present this Newsletter of the Electronics and Communication Engineering Department, Government Engineering College, Patan — a comprehensive documentation of our academic journey, industry exposure, and innovation across the years.

This edition reflects the relentless dedication of our faculty and the enthusiasm of our students. From impactful industry visits and expert lectures to project presentations and faculty development programmes, each activity represents our collective commitment to bridging academia and industry. I express my heartfelt gratitude to all stakeholders — our students, faculty, industry partners, and the college administration — who have continually contributed to making our department a centre of excellence.

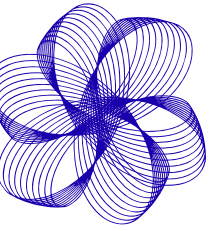
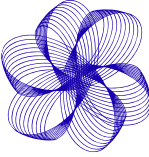
~

Dr. Devendrakumar H. patel

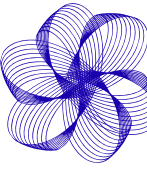
HOD EC Department

Government Engineering College, Patan

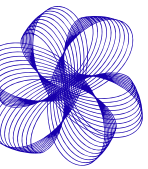




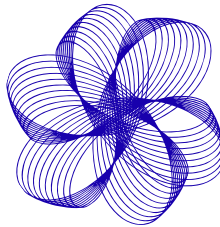
PROGRAM EDUCATIONAL OBJECTIVES (PEOs)



- Accumulate sound technical knowledge that enables to pursue higher education and research.
- Work in multidisciplinary areas with a strong focus on innovation and entrepreneurship.
- To inculcate effective communication skills, managerial skills, as well as professional ethics for successful professional career.

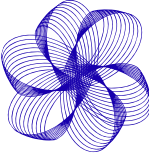


PROGRAM SPECIFIC OUTCOMES (PSOs)

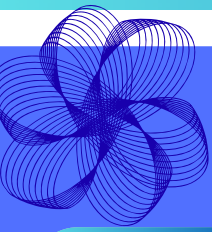


- An ability to apply the conceptual knowledge of Microelectronics, Signal Processing, Microprocessors/Microcontrollers and Communication Systems to solve industrial problems.
- An ability to solve Electronics and communication Engineering problems using the latest hardware and software.





ELECTRONICS AND COMMUNICATION ENGINEERING



Faculty Members



DR. DEVENDRAKUMAR H. PATEL
DESIGNATION: ASSOCIATE
PROFESSOR
AREA OF INTEREST:
MULTIBAND ANTENNAS,
WIRELESS COMMUNICATION



DR. HARESHKUMAR JUDAL
DESIGNATION: ASSISTANT
PROFESSOR
AREA OF INTEREST: DIGITAL
COMMUNICATION, WIRELESS
COMMUNICATION AND DSP



PROF. MIHIRKUMAR PATEL
DESIGNATION: ASSISTANT
PROFESSOR
AREA OF INTEREST: DSP,
COMMUNICATION SYSTEMS



DR. MAULIKKUMAR PATEL
DESIGNATION: ASSISTANT
PROFESSOR
AREA OF INTEREST:
MICROWAVE, ANTENNA,
ELECTRONICS



PROF. JAYESHKUMAR C. PRAJAPATI
DESIGNATION: ASSISTANT
PROFESSOR
AREA OF INTEREST:
COMMUNICATION ENGG,
MICROWAVE ENGG, OPTICAL
FIBER COMMUNICATION



**PROF. SHYAMKUMAR
PANKHANIYA**
DESIGNATION: ASSISTANT
PROFESSOR
AREA OF INTEREST:
EMBEDDED SYSTEM DESIGN,
VLSI TECHNOLOGY



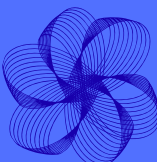
PROF. MEHULKUMAR PATEL
DESIGNATION: ASSISTANT
PROFESSOR
AREA OF INTEREST: VLSI
DESIGN

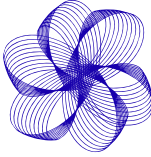


**PROF. ROSHNIBEN
CHAUDHARI**
DESIGNATION: ASSISTANT
PROFESSOR
AREA OF INTEREST:
ANTENNA, RF & MICROWAVE
ENGINEERING, WIRELESS
COMMUNICATION, BASIC
ELECTRONICS



PROF. ANJU VASDEIWANI
DESIGNATION: ASSISTANT PROFESSOR
AREA OF INTEREST: EMBEDDED SYSTEMS,
DIGITAL CIRCUITS AND DESIGNING,
MICROPROCESSOR, MICRO CONTROLLER
& INTERFACING, PROGRAMMING USING C,
BASIC ELECTRONICS





ELECTRONICS AND COMMUNICATION ENGINEERING

FACILITIES OF EC DEPARTMENT

L201 ANALOG & DIGITAL ELECTRONICS LABORATORY

This laboratory is equipped with Analog and Digital Electronics trainers, viz., amplifier, common emitter configuration, common collector configuration, oscillators, multivibrators, power supply, trainers using 555 IC, digital gates, flip flops, counters, and registers kits, etc. Following are few kits:

- Active band pass filter
- Automatic gain control kit
- Butterworth filter trainer
- CE configuration of transistor
- DC motor speed control using SCR



L202 SIMULATION LABORATORY

This laboratory is equipped with the latest computers with advanced simulation software, viz., MATLAB, Multisim, Visual Studio Code, Keil uVision, Atmel Studio and Xilinx etc. Laboratory is equipped with:

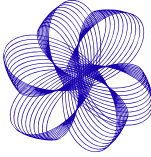
- DIGITAL OSCILLOSCOPE DSOX 2024A
- Workbench Electronics



L-203 ELECTRONIC SYSTEM DESIGN LABORATORY

This laboratory consists of multiple electronics workbenches, having a digital storage oscilloscope, multi meter, function generator, and power supply, which can be helpful in testing and measurements of various electronics circuits or systems.





ELECTRONICS AND COMMUNICATION ENGINEERING

FACILITIES OF EC DEPARTMENT

L211 COMMUNICATION LABORATORY

This laboratory is equipped with Analog and Digital communication trainers, viz., AM, FM, and PM modulation demodulation, superheterodyne receiver, PWM, PPM, PAM, ASK, FSK, PSK kits, Spectrum Analyzer as well as RF and Microwave trainers for microwave communication along with active and passive components. Few trainers are as:

- Amplitude Modulation/Demodulation Trainer
- DSB/SSB AM Receiver
- Fiber Optics Demonstrator
- FM Transmitter Receiver Trainer
- Microwave Bench



L212 PROJECT LABORATORY

This laboratory is designed to provide students with hands-on experience in research and practical applications across various fields of Electronics and Communication Engineering.

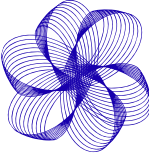


L213 SIGNAL PROCESSING & VLSI LABORATORY

This laboratory consists of various trainers, viz., DSP, FPGA kits, and software required to perform operations on the trainers for Digital Signal Processing, Image Processing, Speech Signal Processing, and VLSI experiments.

- DSP Trainer Kit
- Desktop Computer
(ACER- AMD processor 12th generation)
- Programmable Function Generator, 1 Channel,
50Mhz (AFG3051C)





ELECTRONICS AND COMMUNICATION ENGINEERING

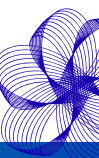
MOOC Sensitization

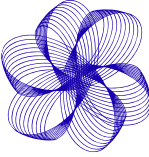
Date: 13th February 2025

Time: 11:00 A.M. onwards

**Venue: Electronics and Communication Engineering Department
(Seminar Hall)**

On 13th February 2025, the EC Department of GEC Patan organized a MOOC Sensitization program for first-year students, coordinated by Prof. S.A. Pankhaniya and Prof. R.A. Chaudhari. The session introduced the concept, benefits, and GTU guidelines of MOOCs, along with relevant courses for various branches. It aimed to promote digital learning and encourage students to explore opportunities beyond the traditional curriculum. An interactive Q&A session helped address student queries and increased interest in MOOCs. The event resulted in greater awareness of online learning platforms and highlighted their role in enhancing academic and career growth.





ELECTRONICS AND COMMUNICATION ENGINEERING

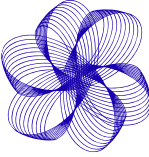
Industrial Visit

Date: 21st February 2025

On 21st February 2025, an industrial visit was organized for 4th and 6th semester students of the EC Department to Alpha numero and Teksun Inc in Ahmedabad. The objective was to provide students with hands-on industrial exposure, particularly in the domains of VLSI Design and Embedded System Design. Experts Mr. Vishnu Vishnav from Alpha-numero and Mr. Birjesh Kamani from Teksun Inc. interacted with students during the visit. A total of 15 students (8 boys and 7 girls) and 2 male faculty members participated.

The visit helped students understand the working environment of industries and introduced them to current trends and innovations in VLSI and embedded systems. They also gained insights into industry expectations and requirements for employment in such sectors. The event was coordinated by Prof. S.D. Joshi and Prof. M.L. Patel, with documentation including feedback and photographs for records.





ELECTRONICS AND COMMUNICATION ENGINEERING

PROJECT PRESENTATION

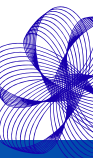
Date: 15th March 2025

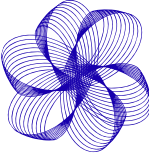
Event Place: EC Department

On March 15th, we participated in a Project Presentation Program organized by the Electronics and Communication Department at Government Engineering College, Patan.

The event featured innovative student projects focused on Arduino Uno-based applications and was attended by our Principal Sir, heads of departments, and senior faculty members. Faculty evaluated each project based on technical execution and presentation, offering valuable insights and encouragement.

We're proud to share that
Meet Mewada secured the first prize, and
Garima Raj won second place.





ELECTRONICS AND COMMUNICATION ENGINEERING

Expert's Talk

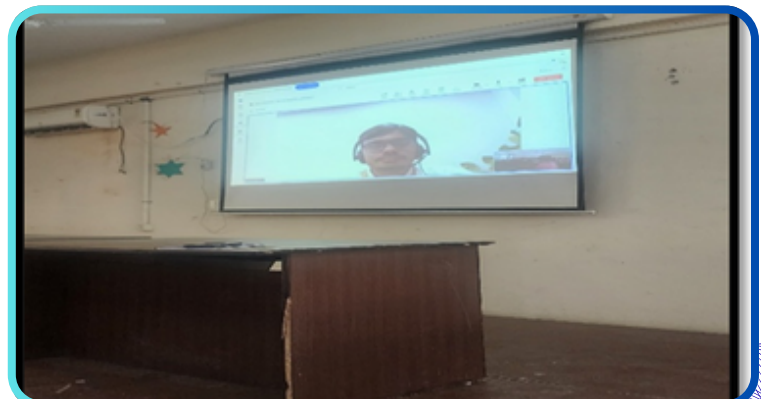
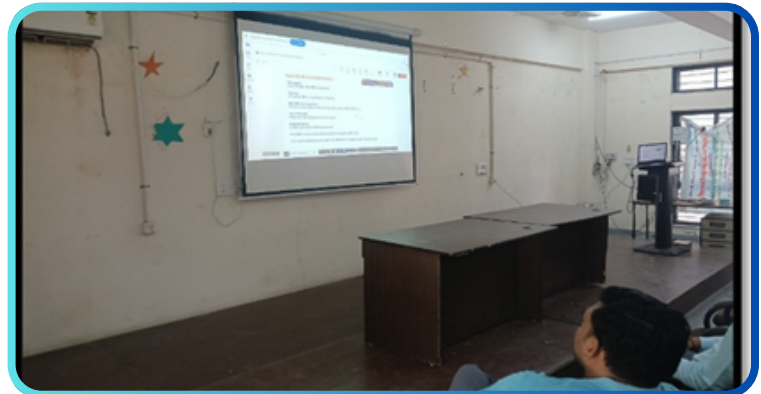
Date: 8th April 2025

Event Place : Electrical Department Seminar Hall

On 8 April, 2025 Electronics And Communication Department Of GEC Patan Organized An Experts Talk On Recent Trends In VLSI Technology. By Hiren R. Soni, Design Verification Technical Lead At Alpha Numero Quest Global Company In Coordination Of Prof. Roshni Chaudhari and Prof. Anju Vasdewani.

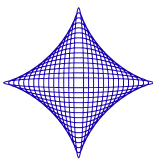
The Virtual Session Was Held In Electrical Department's Seminar Hall At 11 onward.

The Session Provided Valuable Insights Into The Rapidly Evolving Field Of VLSI Technology. Alpha Numero, The Company Behind Hiren R. Soni's Expertise, Specializes In Semiconductor Services.





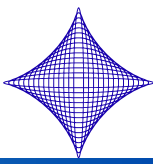
FACULTY ACTIVITIES



ELECTRONICS AND COMMUNICATION ENGINEERING

PROGRAMS PARTICIPATE BY FACULTIES

Sr. No	Title	Name of the faculty	Dept.	Place	Date	Duration
1	INTELLETUAL PROPERTY	Prof. Roshani A. Chaudhari	EC	Online MOOC (IIT Madras)	20/01/2025 to 11/04/2025	12 Weeks
2	Psychology Of Stress, Health And Well-Being	Prof. Maulik B. Patel	EC	Online MOOC (IIT Madras)	20/01/2025 to 11/04/2025	12 Weeks
3	INTELLETUAL PROPERTY	Prof. Jayesh C. Prajapati	EC	Online MOOC (IIT Madras)	20/01/2025 to 11/04/2025	12 Weeks
4	INTELLETUAL PROPERTY	Prof. Hareesh L. Judal	EC	Online MOOC (IIT Madras)	20/01/2025 to 11/04/2025	12 Weeks
5	Embedded system design using ARM	Prof. Anju M. Vasdewani	EC	Online MOOC (IIT Kharagpur)	20/01/2025 to 14/03/2025	8 Weeks
6	Enhancing Soft skills & Personality	Prof. Mihir M. Patel	EC	Online MOOC (IIT Kanpur)	17/02/2025 to 11/04/2025	8 Weeks
7	Enhancing Soft skills & Personality	Prof. Maulik B. Patel	EC	Online MOOC (IIT Kanpur)	17/02/2025 to 11/04/2025	8 Weeks



ELECTRONICS AND COMMUNICATION ENGINEERING

- Prof. R A Chaudhary and Prof. A.M. Vasadevani have Attended One day Webinar on Today's Exams: The Role of Students, Teachers, Parents, and Society on 21/3/2025 by KCG, Education Department, Government of Gujarat



Papers Presented/Published

- Prof. Jayeshkumar C. Prajapati , Prof. Shyam A Pankhaniya, Prof.Kishor G. Maradia has Published a paper on <Enhancing transmission reach: performance analysis of WDM RoF PON OFDM system design with varied QAM schemes= in Journal of Optical Communications De Gruyter January 28, 2025.
- Prof. Shyam A Pankhaniya , Prof. Jayeshkumar C. Prajapati, Prof.Anju M. Vasdewani has Published a paper on IMPLEMENTATION OF SEGMENTED CURRENT STEERING DIGITAL TO ANALOG CONVERTER USING MEMORY LESS PIPELINE DYNAMIC DESIGN TECHNIQUE= in ICTACT Journal on Microelectronics (Volume: 10 , Issue: 4 , Pages: 1945 - 1951) January 2025
- Hitesh Panchal, design "Energy efficient pyramid solar still", design number: 6464827 granted on 06th October 2025 by Intellectual property office, UK.





Sainik Welfare Fund

*Event date : 19th December to 21st March
Venue : Patan City*

The Sainik Welfare Fund initiative at Government Engineering College, Patan, in 2025, showcased the collective spirit of students and faculty towards honoring the sacrifices of ex-servicemen and the families of martyred soldiers. Guided by their professors and led by the NSS unit, volunteers actively collected donations within the college, receiving full support from Principal Dr. B.J. Shah and all departments. The activity not only strengthened students' values of service, teamwork, and patriotism but also extended beyond the campus, reaching government offices like the Collector Office, Mamlatdar Office, Nagarpalika, Police Stations, and several schools, where they spread awareness and gathered widespread support from officers, staff, and the general public.

Through their united efforts, the NSS volunteers raised an impressive ₹1,31,131 — the highest amount collected by the college in nine years and the top contribution among educational institutions in Gujarat State. The fund was formally submitted to the Vice-Chancellor of Gujarat Technological University (GTU) at Chandkheda, with GEC Patan proudly recognized for making the largest contribution among all NSS units under GTU. This achievement reflected the deep sense of patriotism, dedication, and teamwork among the students, leaving a lasting impact and inspiring continued commitment to serving the nation.





Tobacco Free Gujarat

Event Date : 8th January 2025

Venue : GEC Patan Campus



On 8th January 2025, the NSS Unit of GEC Patan, in collaboration with the Taluka Health Office, successfully organized a seminar on the theme 8Tobacco-Free Gujarat9 at

Government Engineering College, Patan. The event aimed to spread awareness about the harmful effects of tobacco and promote a healthy lifestyle.



Eminent guests Dr. Ruchit Patel and Dr. Jinal Sarvi from Kungher Health Center addressed the students, along with the college principal Dr. B. J. Shah, who motivated attendees to embrace a tobacco-free life. The seminar began with a Deep Pragatya ceremony and the NSS Song, highlighting the values of the NSS. An elocution competition was also held as part of the event, where students actively shared their thoughts on tobacco's impact and healthy living. Winners and participants were felicitated with prizes and certificates, marking a successful conclusion to the awareness initiative.



Cyber Awareness

Event Date : 2nd April 2025

Venue : Computer Science Engineering Seminar Hall

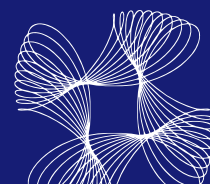


The NSS unit of Government Engineering College, Patan, successfully conducted a Cyber Awareness Program to educate students and faculty on the importance of cyber safety.

The event saw enthusiastic participation from various departments and featured expert-led sessions covering topics like phishing, password protection, social media safety, and data privacy.



Respected guests from the SP Office, including PSI S.K. Desai, PC Nikunj Sir, and Rajni Ma'am, graced the occasion. Interactive discussions, real-life cybercrime case studies, and Q&A sessions helped attendees understand the significance of digital safety. The program concluded with a pledge by students and staff to adopt and promote safe online practices.





COMMITTEE ACTIVITIES

NATIONAL SERVICE SCHEME



World heritage day

Event Date : 18th April 2025

Venue : Patan and Modhera city

On World Heritage Day, a vibrant Modhera Heritage Walk was organized under the "Vikas Bhi, Virasat Bhi" initiative by the Government of India, in collaboration with NSS,

Invincible NGO, and ASI. Around 100 youth, including 20 students from GEC Patan led by Prof. Naran Dabhi Sir, actively participated.

The walk explored key heritage sites in Patan and Modhera like Rani ki Vav, Patola House, Chhatrapateshwar Temple, Jain Temple, and Modhera Sun Temple. Cultural music programs celebrated Gujarat's rich tradition. A Youth Roundtable Conference was held at Modhera Sun Temple featuring eminent personalities like Mrs. Pabiben Rabari, Mr. Paresh Rathwa, and Dr. Subha Majumdar, encouraging youth involvement in heritage conservation and cultural innovation.





EDITORIAL TEAM

Patron

Dr. D. H. Patel

HOD, Electronics and Communication Department

Editors-in-Chief

Dr. M. B. Patel - Electronics & Communication
Engineering

Content & Design Student Team

Joshi Parv Kashyap (240220111004 - EC)

Verma Samir Santoshbhai (240220111033 - EC)

Dahivat Ahir (250022011101 - EC)

Shiv Patel (250220111023 - EC)

Harsh Prajapati (250220111025 - EC)

Rushang Thakar (250220111034 - EC)